

National University of Computer and Emerging Sciences, Lahore Campus

Course:	Design and Analysis of Algorithms
Program:	BS(Data Science)
Duration:	25 Minutes
Paper Date:	24-Oct-2023
Section:	5B
Exam:	Quiz 2

Course Code:	CS2009
Semester:	Fall 2023
Total Marks:	10
Weight	%
Page(s):	4

Q1) Consider the following recursive formula:

$$C_0 = 1 \text{ and } C_{n+1} = \sum_{i=0}^n C_i C_{n-i} \text{ for } n \geq 0$$

For example $n = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 \dots$ output : 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, ...

Write an iterative dynamic programming solution for the above recurrence. What is its time complexity? [5 Marks]

Name: _____

Reg #: _____

Section: _____

Q2) Dry run binary knapsack dynamic programming algorithm on the following problem:Capacity = $W = 8$

Items	1	2	3	4	5
Weights	4	2	2	7	3
values	3	1	3	7	5

Fill the following table. Place arrows pointing to the previous cell which is used for defining values of current cell. [1 + 4 = 5 Marks], 1 mark for writing correct values, 4 marks for placing correct arrows. Also print the items which are used for finding optimal solution.

Item weight	0	1	2	3	4	5
0						
1						
2						
3						
4						
5						
6						
7						
8						

0-1 Knapsack Algorithm

for $w = 0$ to W

$B[0,w] = 0$

for $i = 0$ to n

$B[i,0] = 0$

 for $w = 0$ to W

 if $w_i \leq w$ // item i can be part of the solution

 if $b_i + B[i-1, w-w_i] > B[i-1, w]$

$B[i, w] = b_i + B[i-1, w-w_i]$

 else

$B[i, w] = B[i-1, w]$

 else $B[i, w] = B[i-1, w]$ // $w_i > w$